

METHODOLOGY & VALIDATION PROTOCOL | *Pharmaceutical Analysis & Method Validation*

RP-HPLC Method Development and Validation for Simultaneous Estimation of Metformin and Remogliflozin in Combined Solid Dosage Forms

Authors: A. K. Srivastava, Vivek Sharma, N. Gupta, P. M. Joshi • Corresponding: Vivek Sharma (pharmionex.journal@gmail.com)

ABSTRACT

Reversed-phase isocratic HPLC method validated strictly under ICH Q2(R1) with recovery rates between 99.82% and 100.14%.

Keywords: RP-HPLC; Metformin; Remogliflozin; ICH Q2(R1); Validation; Stability-Indicating.

1. SCIENTIFIC INTRODUCTION & RATIONALE

This manuscript addresses critical empirical challenges in contemporary pharmaceutical formulation, method validation, and clinical translation. Experimental procedures were conducted strictly under Good Laboratory Practice (GLP) and standard regulatory frameworks.

2. METHODOLOGICAL DATA & COMPLIANCE

Assay Parameter / Quality Metric	Measured Value	Standard Benchmark	Status
Assay Content Uniformity	99.82 ± 0.45%	98.0% – 102.0%	Complies
Precision (% RSD)	0.82%	< 2.0%	Complies
Plagiarism Screening (Turnitin)	4.8%	< 10.0%	Complies

3. CONCLUSION

The findings establish reproducible methodologies contributing directly to scientific advancement and patient healthcare.

DECLARATIONS: Institutional funding support; no competing financial interests declared; datasets available upon request.